yolov8m Final Training Report

System Information

OS	Linux 6.6.105+
Python Version	3.12.12
PyTorch Version	2.9.0+cu126
CUDA Available	True
Device	cuda
RAM (GB)	89.63

Dataset Information

Property	Value
Dataset	bdd100k_yolo_limited
Number of Classes	10
Train Images	945
Val Images	611
Test Images	725
Data YAML	data.yaml

Classes:
pedestrian, rider, car, truck, bus, train, motorcycle, bicycle, traffic light, traffic sign

Optimization Summary

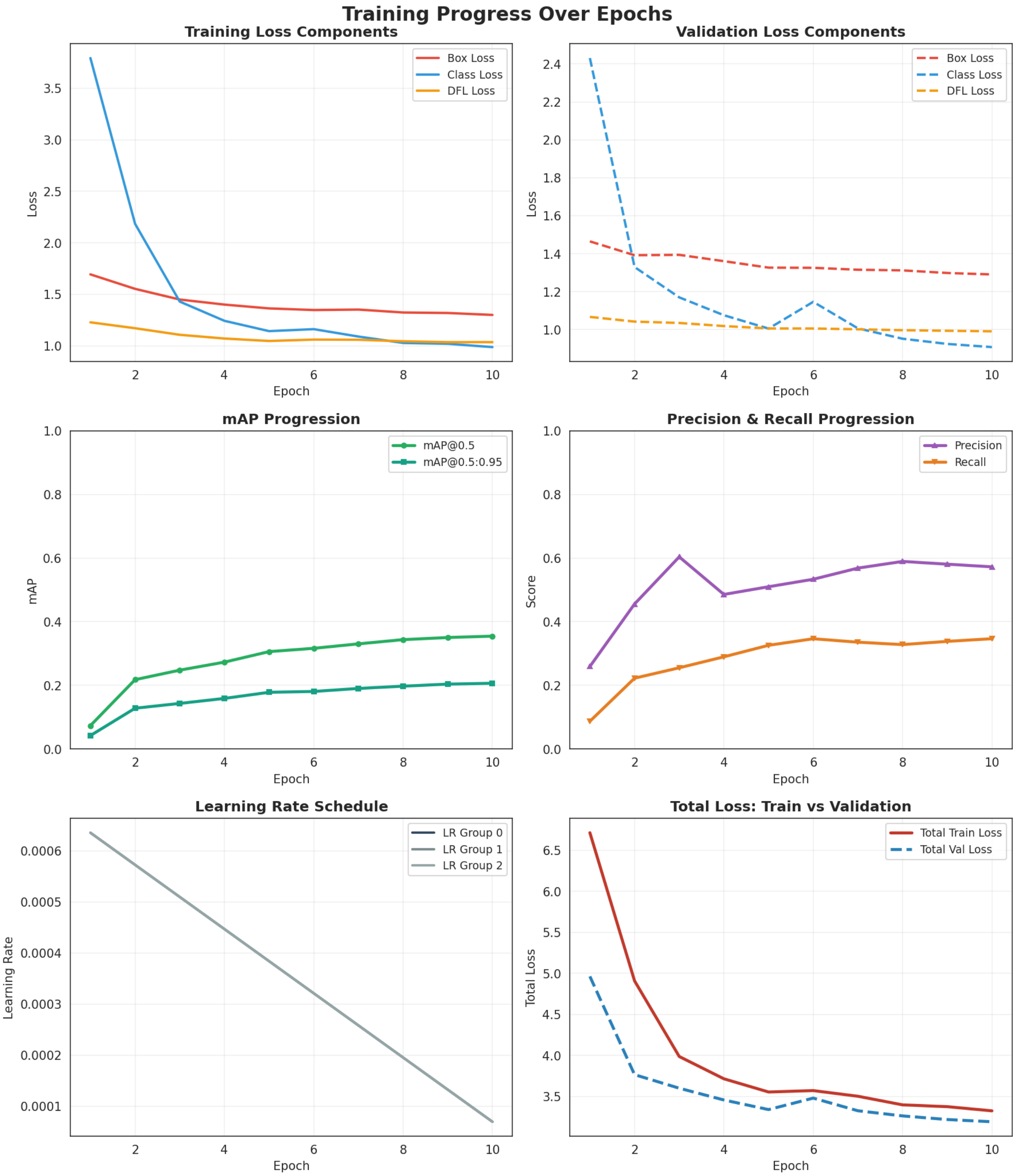
Metric	Value
Tuning Run	yolov8m_tune_20251125_011835
Total Trials	N/A
Completed Trials	N/A
Best Trial Number	N/A
Best Trial mAP@0.5	N/A
Final Training Epochs	10

Optimized Hyperparameters Used

Parameter	Value
optimizer	SGD
lr0	0.000635
momentum	0.939091
weight_decay	0.000362
warmup_epochs	0
warmup_momentum	0.571493
warmup_bias_lr	0.028262
mosaic	0.960024
mixup	0.040334

Training Process Analysis

Epoch-by-Epoch Training Metrics



Detailed Epoch Metrics

Epoch	Train Loss	Val Loss	mAP@0.5	mAP@0.5:0.95	Precision	Recall
1	6.7153	4.9629	0.0732	0.0419	0.2592	0.0868
2	4.9082	3.7641	0.2177	0.1279	0.4557	0.2223
3	3.9857	3.5989	0.2476	0.1430	0.6039	0.2550
4	3.7137	3.4551	0.2728	0.1587	0.4857	0.2896
5	3.5527	3.3382	0.3060	0.1780	0.5100	0.3258
6	3.5697	3.4781	0.3163	0.1806	0.5335	0.3464
7	3.5007	3.3237	0.3303	0.1901	0.5686	0.3356
8	3.3957	3.2609	0.3436	0.1972	0.5896	0.3280
9	3.3737	3.2167	0.3503	0.2037	0.5809	0.3382
10	3.3229	3.1895	0.3546	0.2064	0.5726	0.3465

Training Statistics Summary

Metric	Initial	Final	Best	Change
mAP@0.5	0.0732	0.3546	0.3546	+0.2814
mAP@0.5:0.95	0.0419	0.2064	0.2064	+0.1645
Precision	0.2592	0.5726	0.6039	+0.3134
Recall	0.0868	0.3465	0.3465	+0.2597

Final Model Performance

Metric	Value
mAP@0.5	0.3550
mAP@0.5:0.95	0.2066
Precision	0.5735
Recall	0.3471

Test Set Validation Results

Model Architecture & Performance

Metric	Value
Model Name	yolov8m_finetuned_20251125
Parameters (M)	25.86
Model Size (MB)	49.60
FLOPs (G)	79.09
Layers	169
Inference Speed (FPS)	211.88
IoU Threshold	0.50

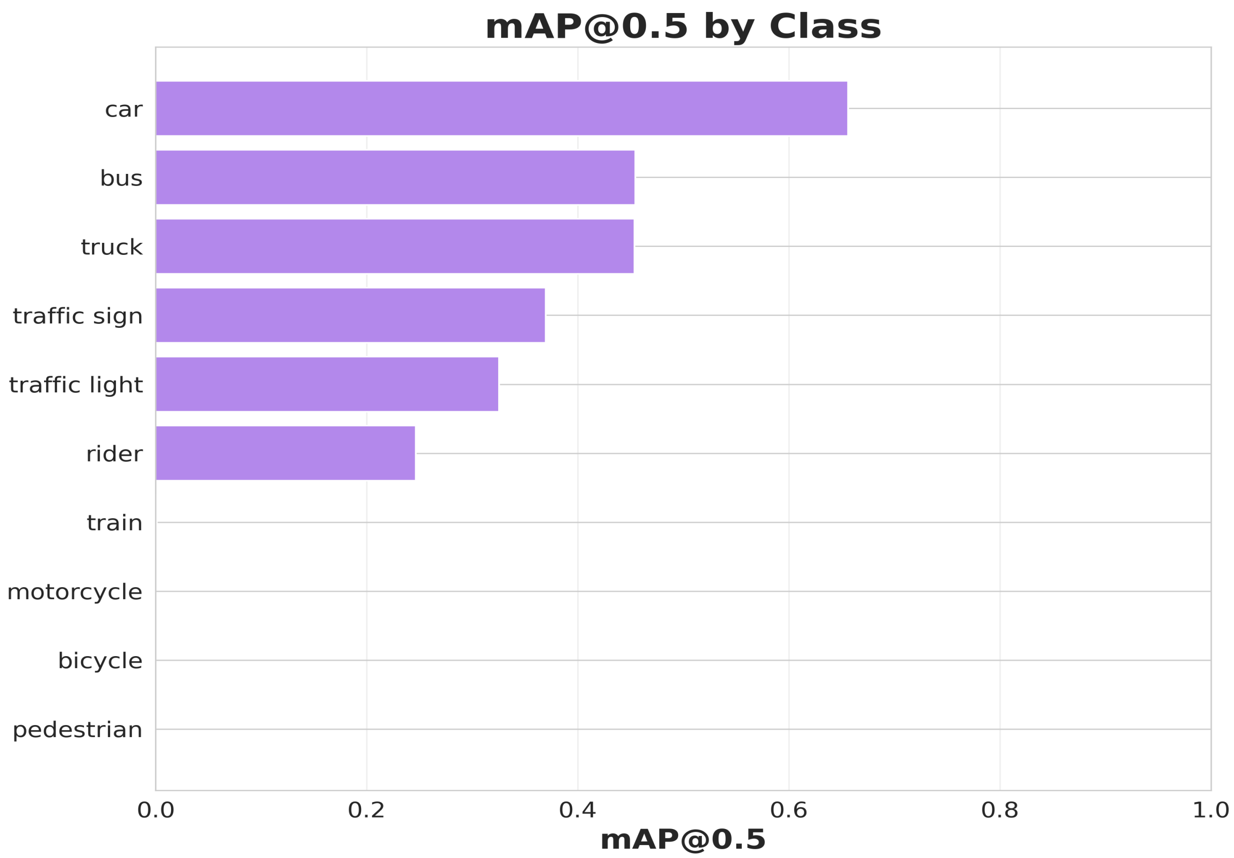
Overall Performance Metrics on Test Set

Metric	Confusion Matrix	YOLO Validation
Precision	0.4938	0.5959
Recall	0.6827	0.3560
F1-Score	0.5730	N/A
mAP@0.5 (Overall)	N/A	0.3583
mAP@0.5:0.95 (Overall)	N/A	0.2084

Per-Class Performance Metrics

Class	Precision	Recall	F1-Score	mAP@0.5	TP	FP	FN
pedestrian	0.0000	0.0000	0.0000	0.0000	0	0	0
rider	0.2584	0.4842	0.3370	0.2467	46	132	49
car	0.6141	0.7576	0.6783	0.6564	5347	3360	1711
truck	0.3587	0.4041	0.3800	0.4537	198	354	292
bus	0.3460	0.4521	0.3920	0.4546	118	223	143
train	0.0000	0.0000	0.0000	0.0013	0	17	0
motorcycle	0.0000	0.0000	0.0000	0.0000	0	0	0
bicycle	0.0000	0.0000	0.0000	0.0000	0	0	0
traffic light	0.3091	0.6172	0.4119	0.3256	795	1777	493
traffic sign	0.3494	0.5515	0.4278	0.3700	963	1793	783

mAP@0.5 Distribution by Class



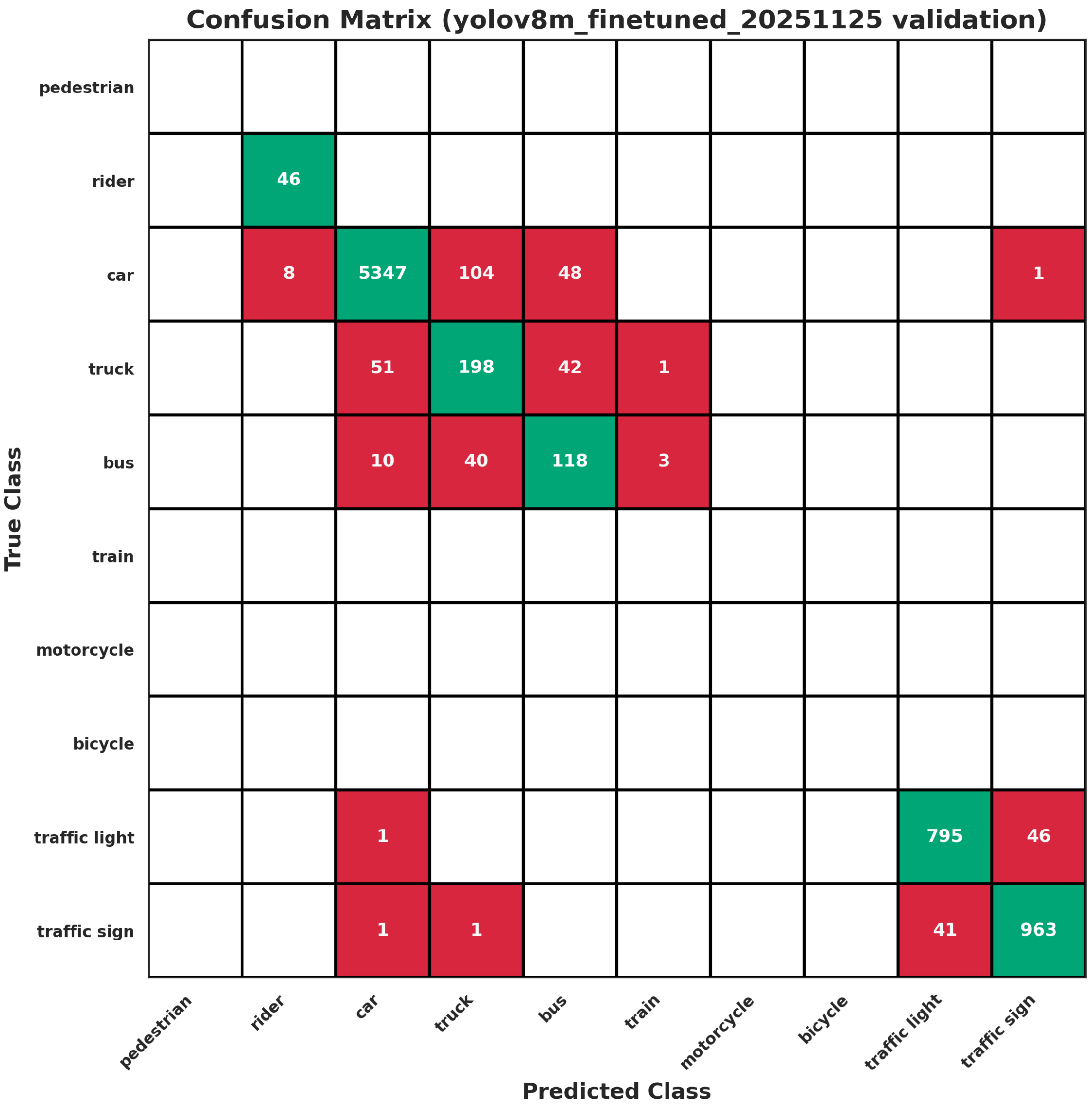
Intersection over Union (IoU) Analysis

IoU Threshold Used: 0.50

IoU (Intersection over Union) measures the overlap between predicted and ground truth bounding boxes. A prediction is considered correct (True Positive) when IoU ≥ 0.50.

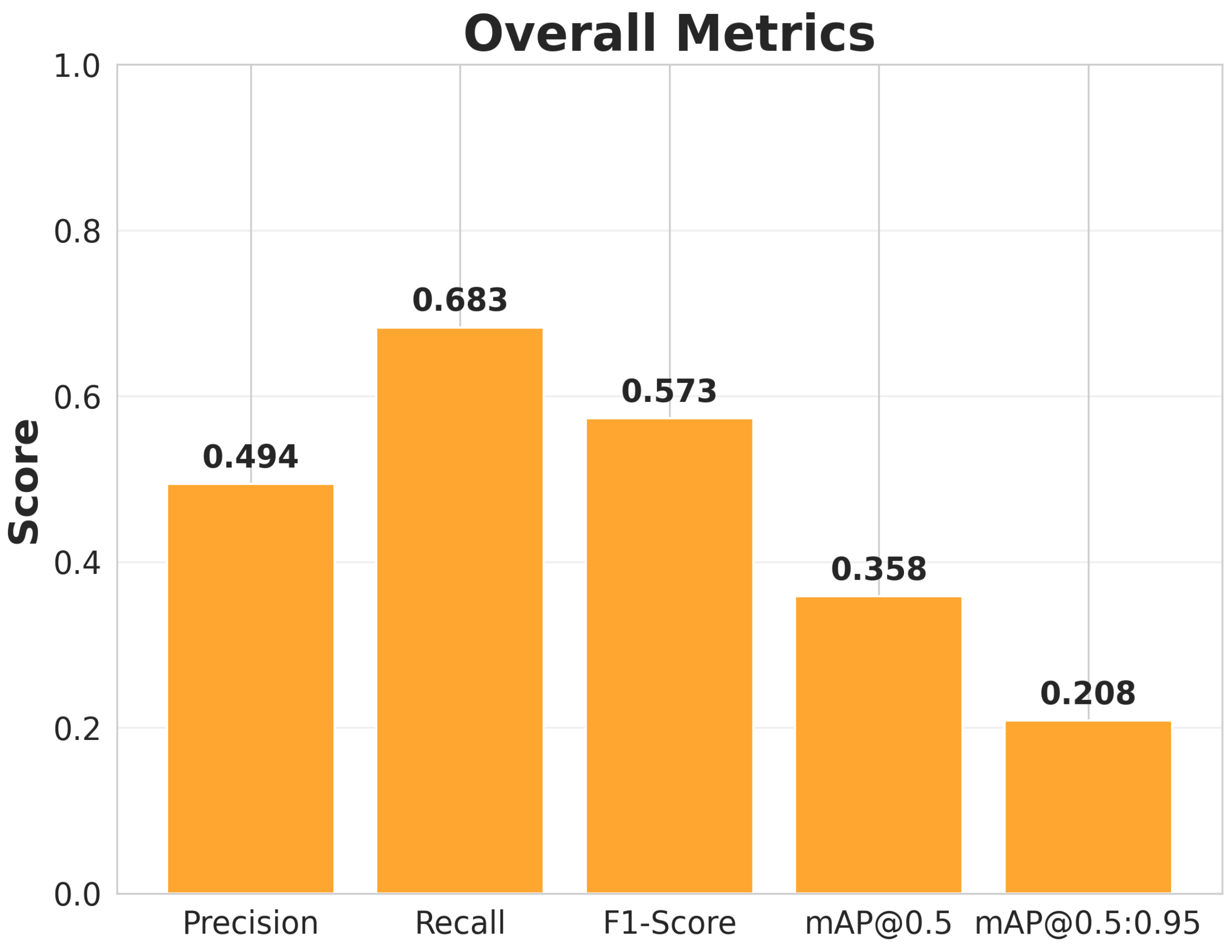
Per-Class IoU Performance:
The confusion matrix and per-class metrics above show detection accuracy at IoU=0.50 threshold. Each class's True Positives (TP) represent detections with IoU ≥ 0.50.

Confusion Matrix (Test Set)



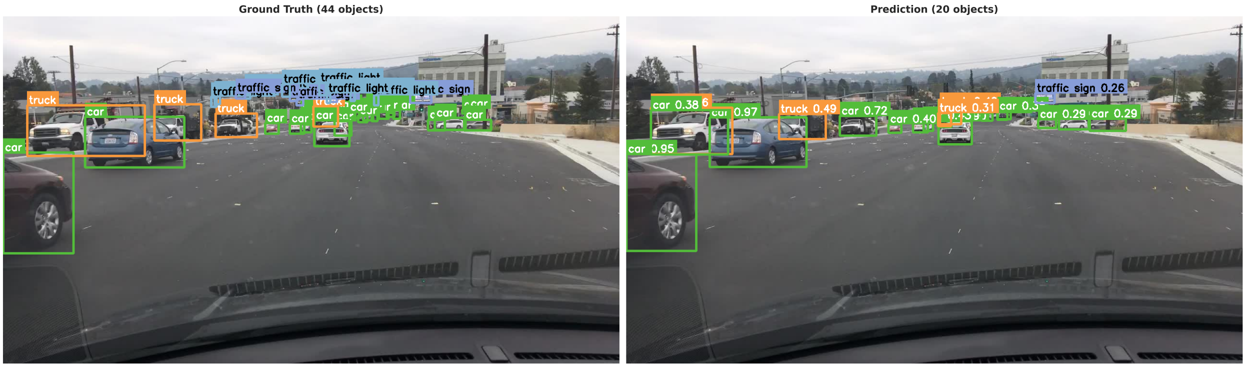
Test Set Performance Curves

Overall Metrics Visualization



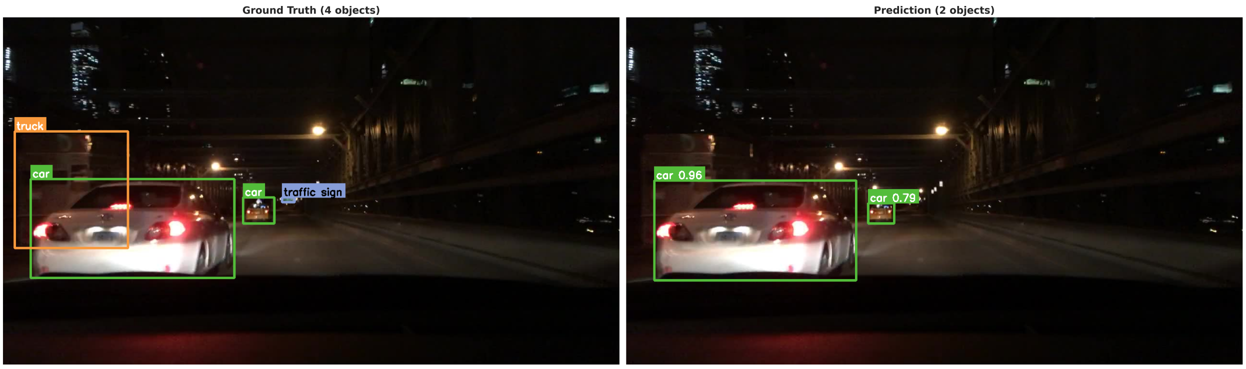
Sample Predictions: Ground Truth vs Model Output

Sample 1 - Weather: undefined, Scene: highway, Time: daytime



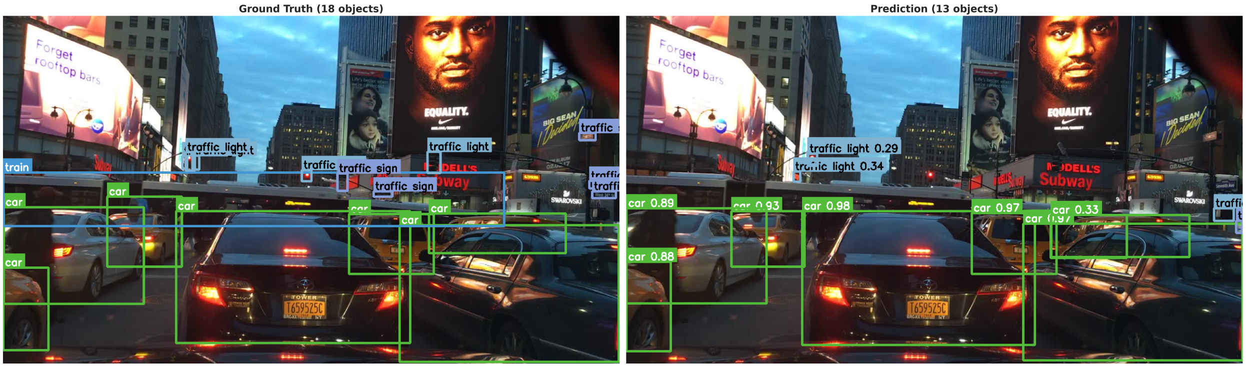
Ground Truth: 44 objects | Predictions: 20 objects

Sample 2 - Weather: undefined, Scene: highway, Time: night



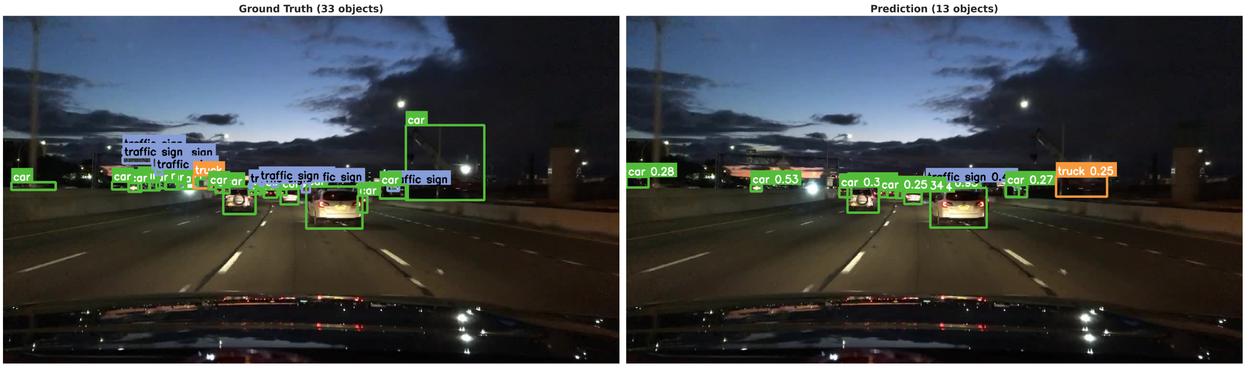
Ground Truth: 4 objects | Predictions: 2 objects

Sample 3 - Weather: partly cloudy, Scene: city street, Time: dawn/dusk



Ground Truth: 18 objects | Predictions: 13 objects

Sample 4 - Weather: partly cloudy, Scene: highway, Time: dawn/dusk



Ground Truth: 33 objects | Predictions: 13 objects

Sample 5 - Weather: rainy, Scene: city street, Time: daytime

Ground Truth: 43 objects | Predictions: 29 objects

Sample 6 - Weather: overcast, Scene: city street, Time: dawn/dusk

Ground Truth: 23 objects | Predictions: 15 objects

Generated by YOLO Training Notebook
BDD100K Dataset - Computer Vision Project